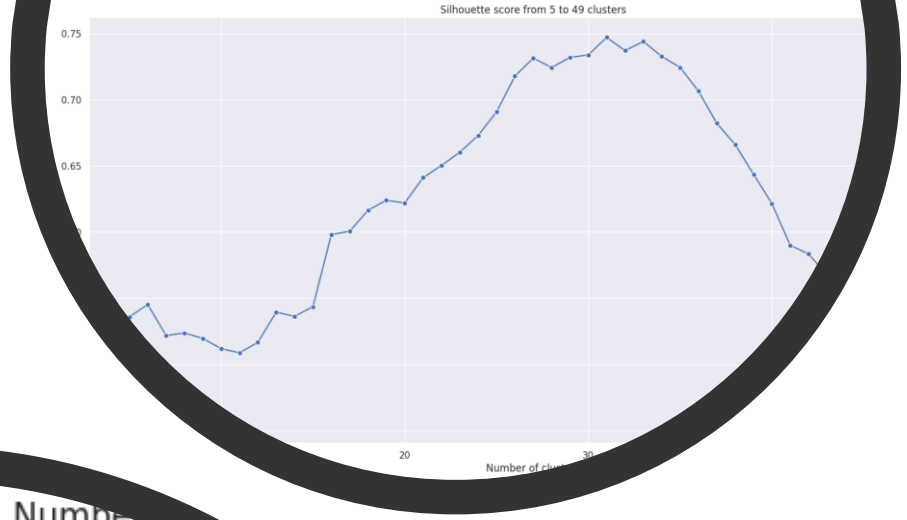
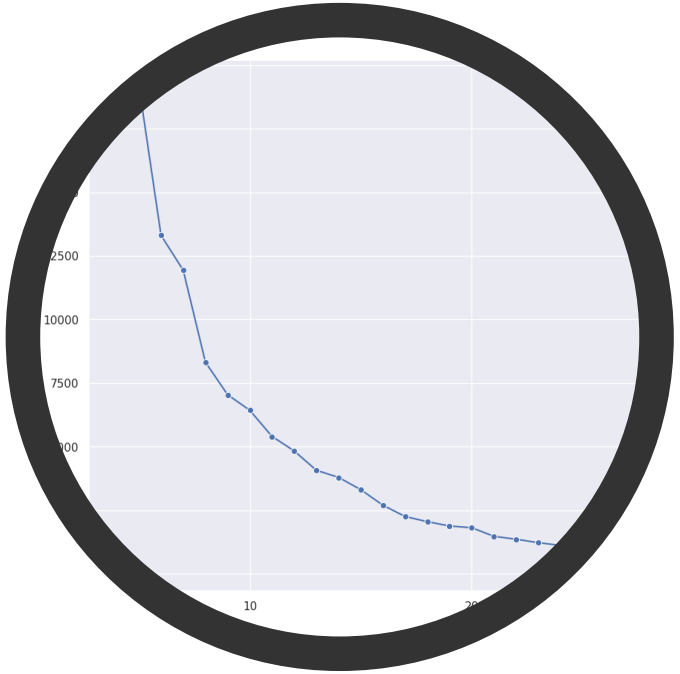
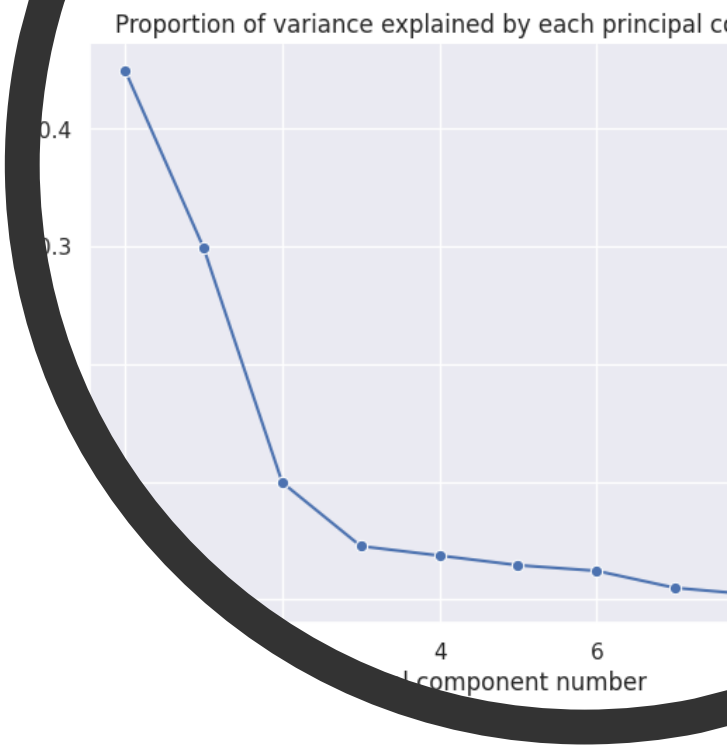
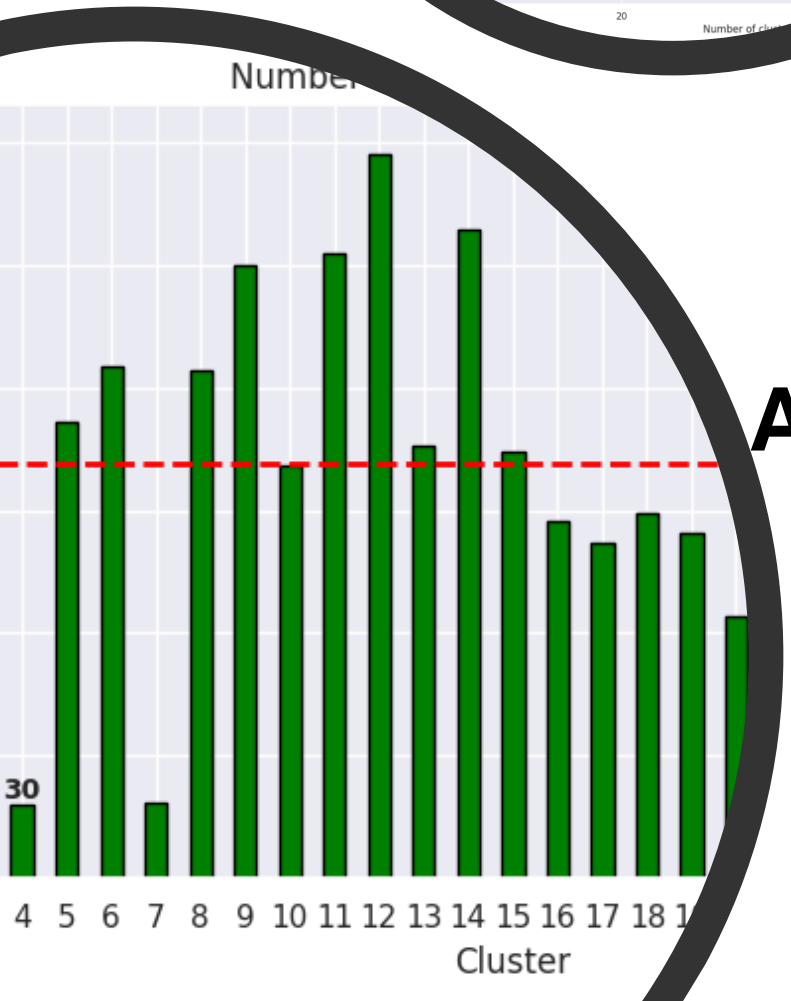




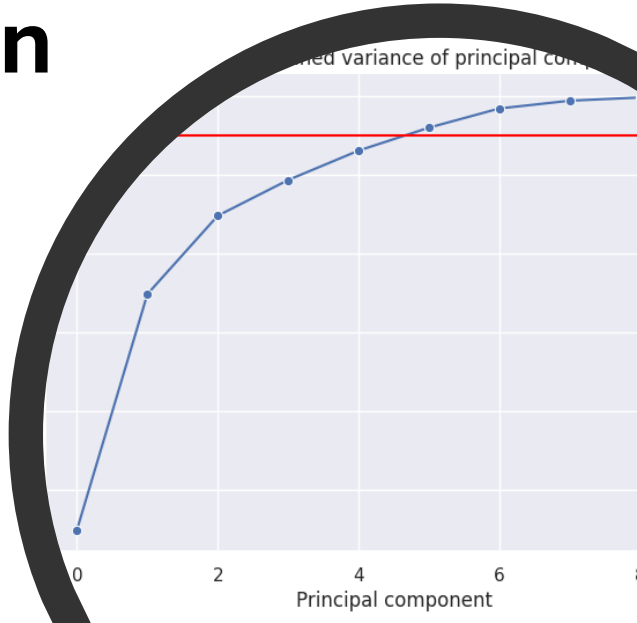




Automated Playlist Creation

Moosic

Leyla Bakacak Karabenli Ciba Baharestani





Mission

- **Are Spotify's audio features able to identify "similar songs", as defined by humanly detectable criteria?**

Partially!

- **Is K-Means a good method to create playlists?**

Yes, but !

Features Selection

Numerical (to keep)

- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence
- Tempo
- Duration_ms

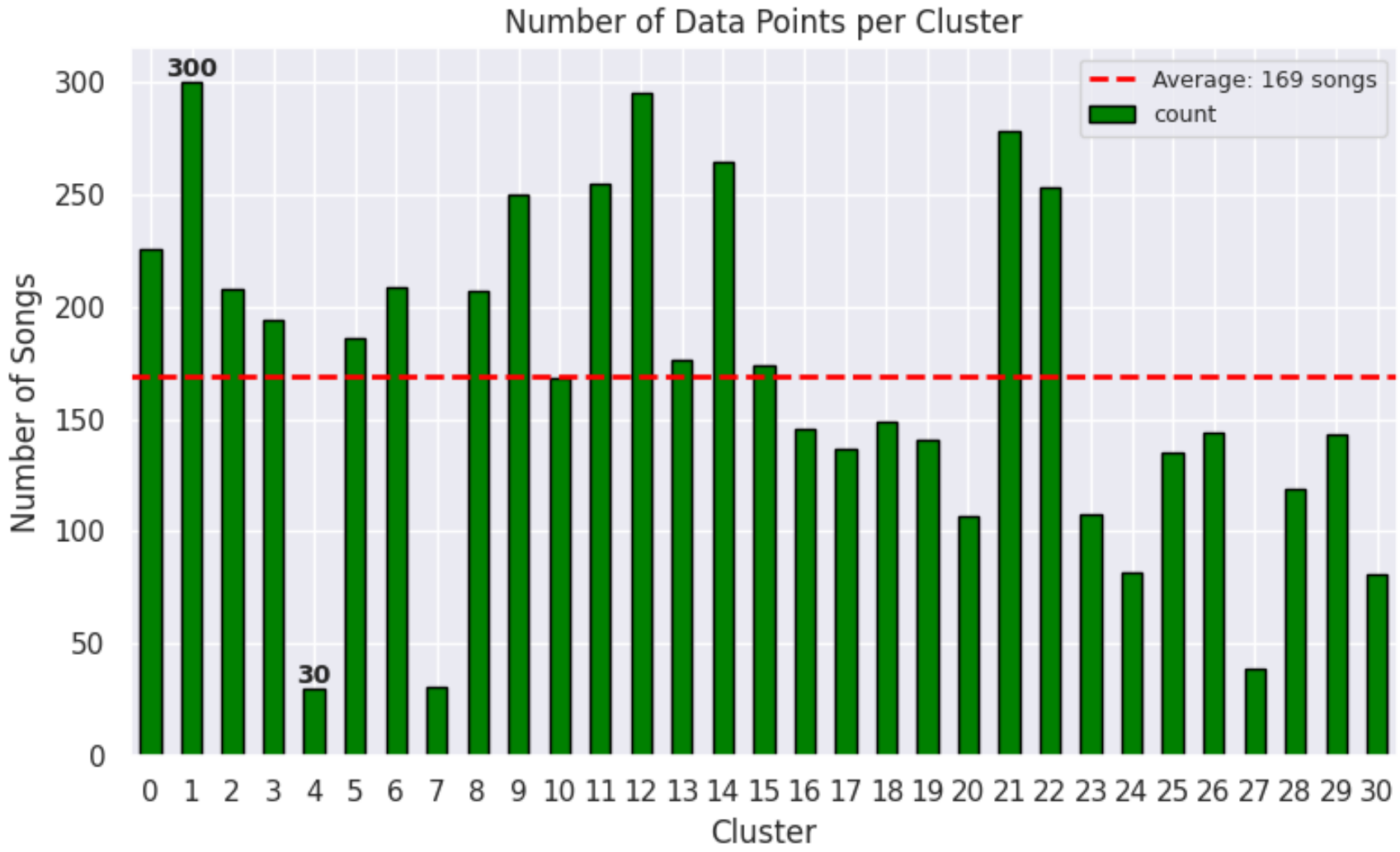
Numerical (to drop)

- Mode
- Time_signature
- Key

Categorical (to drop)

- Artist
- Name
- ID
- HTML

Using K-Means



Do the clusters make musical sense?

The largest cluster

Artist	Song Name	Genre Info
Coldplay	Paradise (Tiësto Remix)	Electronic / EDM / Progressive House
Beatrice Egli	Mein Herz	German Schlager / Pop-Schlager
Houston	I Like That	2000s R&B / Pop-Rap
Sonny & The Sunsets	Icelene's Loss	Indie Pop / Garage Pop
Plan B	Chica Ven Pa' Aca	Reggaeton / Latin Urban
Wisin & Yandel	Llamé Pa' Verte	Classic Reggaeton / Latin Urban
Calvin Harris (with Dua Lipa)	One Kiss	Dance-Pop / Deep House
Maroon 5	Payphone	Pop / Pop-Rock
Love Inks	Blackeye	Indie Pop / Minimal Synth-Pop
Rihanna	We Found Love	Dance-Pop / Electro House

The smallest cluster

Artist	Song Name	Genre Info
Dave Brubeck	Besame Mucho	Cool Jazz
Pedro Aznar	Zamba Del Carnaval	Argentine Folk / Trova
Giuseppe Verdi	Il Trovatore: Vedi! le fosche...	Classical Opera / Romantic Era
The Isley Brothers	Shout, Pts. 1 & 2	1950s R&B / Soul / Rock & Roll
Gilberto Gil	Back In Bahia (Ao Vivo)	MPB (Música Popular Brasileira) / Tropicalia
Elomar, Geraldo Azevedo, Vital Farias, Xangai	Sabor Colorido / Moça Bonita	Regional Brazilian Folk / Violão Acoustic
Keith Jarrett	One For Majid - Live	Avant-Garde Jazz / Post-Bop
Léo Delibes	Lakmé: Dôme épais... (The Flower Duet)	Classical Opera / French Romanticism
Gilberto Gil	Se eu quiser falar com Deus (Studio & Live)	MPB / Spiritual Acoustic Folk

Do the clusters make musical sense?

The Average Cluster

Artist	Song name	Genre
Angelcorpse	Into The Storm Of Steel	Blackened Death Metal
Necrophobic	Unholy Prophecies	Blackened Death Metal / Swedish Death Metal
Viogression	Maggot Synod	Old-School Death Metal
Therion	Asphyxiate with Fear	Early Death Metal / Symphonic Metal (Early era)
Cemetary	Nightmare Lake	Gothic Metal / Death-Doom Metal
Asphyx	Frozen Soul	Death-Doom Metal
Unleashed	To Asgard We Fly	Viking Metal / Death Metal
Tchami	After Life (feat. Stacy Barthe)	Future House / Electronic Dance Music (EDM)
Devastation	Freewill	Thrash Metal
Fleshcrawl	Phrenetic Tendencies	Old-School Death Metal

Is K-Means a good method?

Advantages	Disadvantages
Easy to Apply	Must Choose "K" Manually
Easy to Understand and Explain	Categorical Features
Revealing Hidden Patterns	Needing Expert Human Monitoring
Scalable	
Fast	
Stable Grouping Solution	

How we make playlists better



1. Keep experts in the loop

Use clusters as draft playlists. Ask music experts to review, rename and refine them.



2. Semi-supervised ML



3. Learn from listeners

Track skips, saves, likes, playlist follows and songs users remove or add.



4. Add richer context

Test release year, language and lyric themes (e.g. love, miss) where available and appropriate.



THANK YOU!